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(54) **CRYSTALLINE FORMS OF OZANIMOD AND OZANIMOD HYDROCHLORIDE, AND PROCESSES FOR PREPARATION THEREOF**

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(22) Filed: **Jan. 21, 2020**

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Related U.S. Application Data

(63) Continuation of application No. 16/310,328, filed as application No. PCT/CN2017/088314 on Jun. 14, 2017, now Pat. No. 11,111,223.

(30) **Foreign Application Priority Data**

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Jun. 29, 2016 (CN) 201610493910.7

(51) **Int. Cl.**
C07D 271/06 (2006.01)

(52) **U.S. Cl.**
CPC ***C07D 271/06*** (2013.01); *C07B 2200/13* (2013.01)

(58) **Field of Classification Search**
CPC C07D 271/06
See application file for complete search history.

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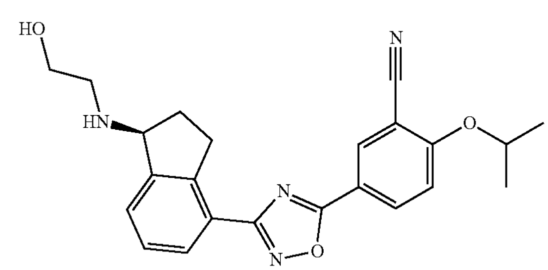
Primary Examiner — Matthew P Coughlin
(74) *Attorney, Agent, or Firm* — McCarter & English, LLP; Wei Song

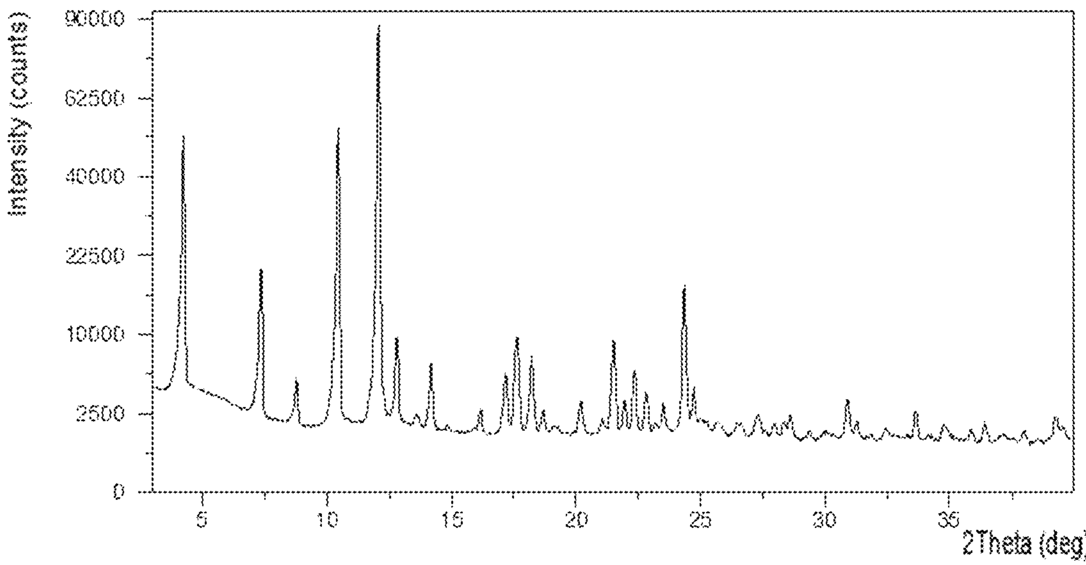
(57) **ABSTRACT**

The present disclosure is directed to novel crystalline forms of ozanimod and ozanimod hydrochloride, as well as preparation method thereof. Said crystalline forms of ozanimod and ozanimod hydrochloride can be used for treating autoimmune diseases, particularly used for preparing drugs for treating multiple sclerosis and ulcerative colitis. The crystalline forms of the present disclosure have one or more advantages in solubility, melting point, stability, dissolution, bioavailability and processability and provide new and better choices for the preparation of ozanimod drug product, and are very valuable for drug development.

(I)



7 Claims, 25 Drawing Sheets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22500
10000
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20
25
30
35
2Theta (deg)

FIG.1

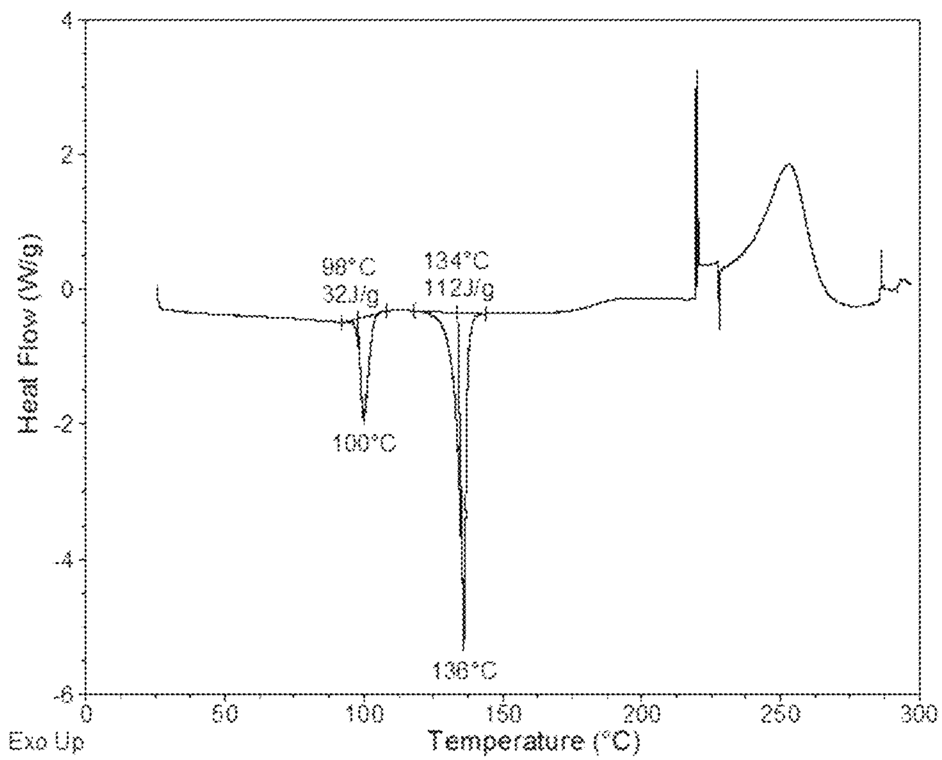

Heat Flow (W/g)
4
2
0
-2
-4
-6
98°C
32J/g
100°C
134°C
112J/g
136°C
0
50
100
150
200
250
300
Exo Up
Temperature (°C)

FIG.2

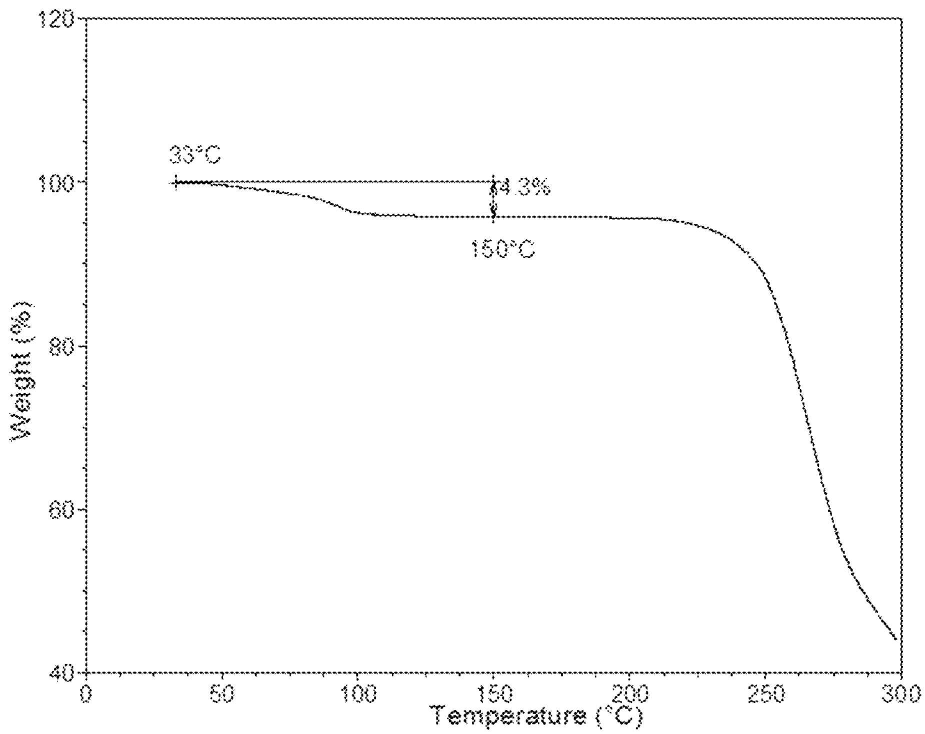
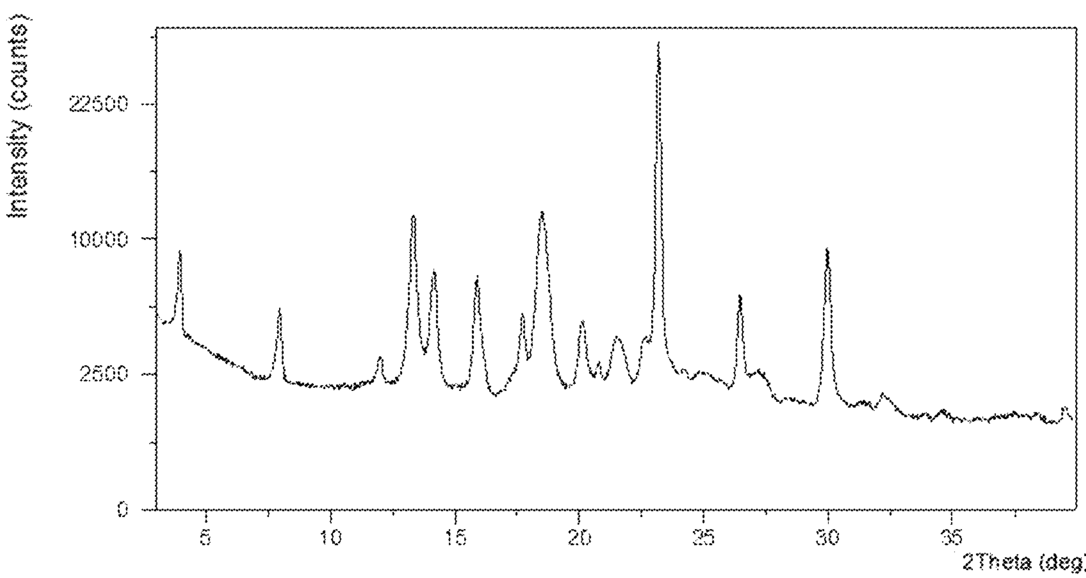

33°C
4.3%
150°C
Weight (%)
Temperature (°C)
120
100
80
60
40
0
50
100
150
200
250
300

FIG.3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30
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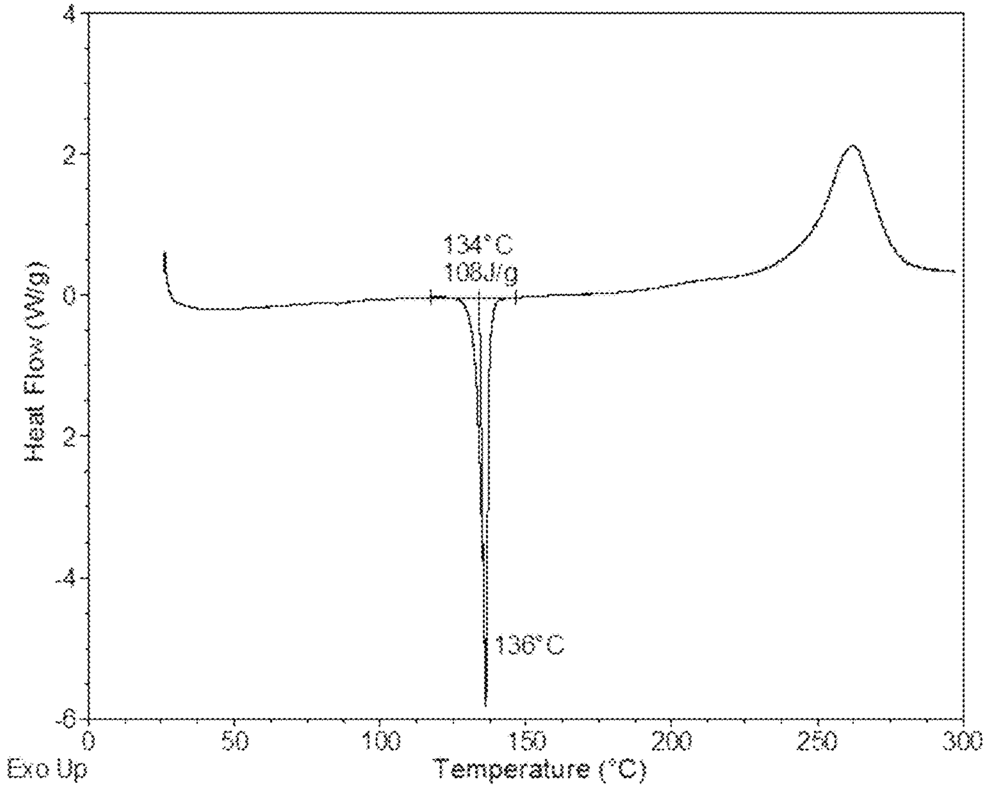
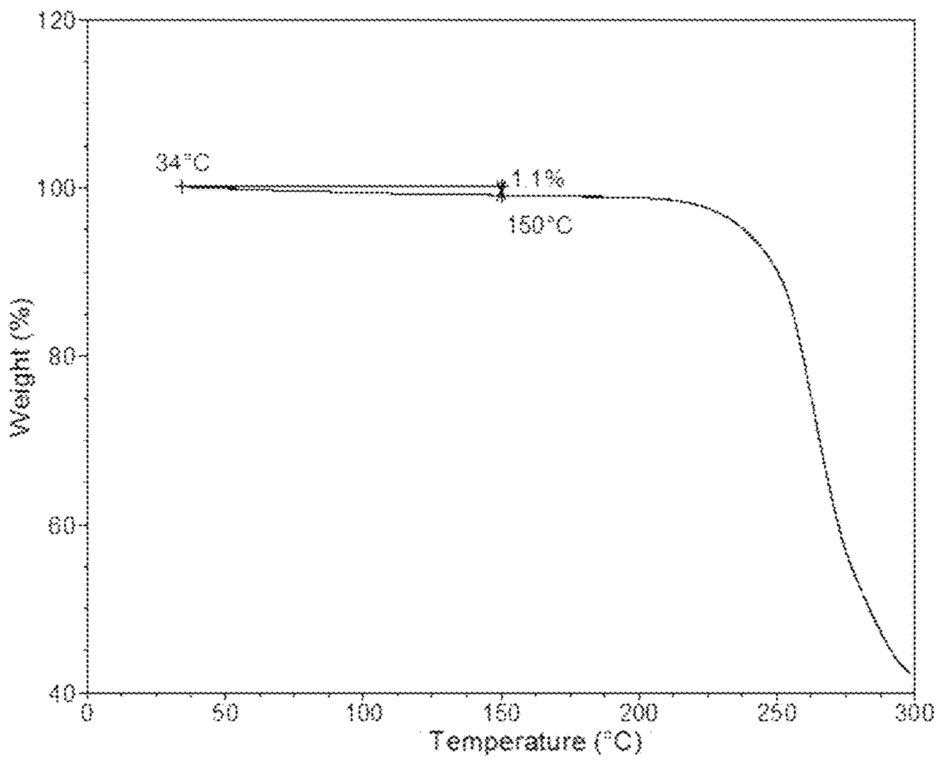
2Theta (deg)

FIG.4



134°C
108J/g
136°C
Heat Flow (W/g)
Exo Up
Temperature (°C)

FIG.5



34°C
1.1%
150°C
Weight (%)
Temperature (°C)

FIG.6

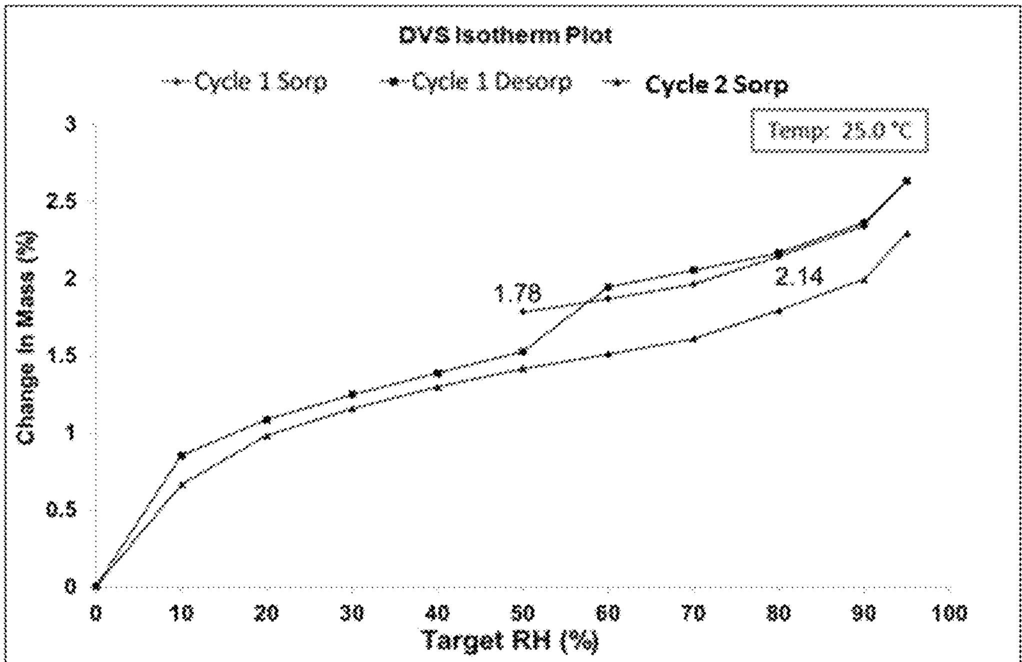

DVS Isotherm Plot
Cycle 1 Sorp
Cycle 1 Desorp
Cycle 2 Sorp
Temp: 25.0 °C
Change In Mass (%)
Target RH (%)
1.78
2.14
0
0.5
1
1.5
2
2.5
3
0
10
20
30
40
50
60
70
80
90
100

FIG.25

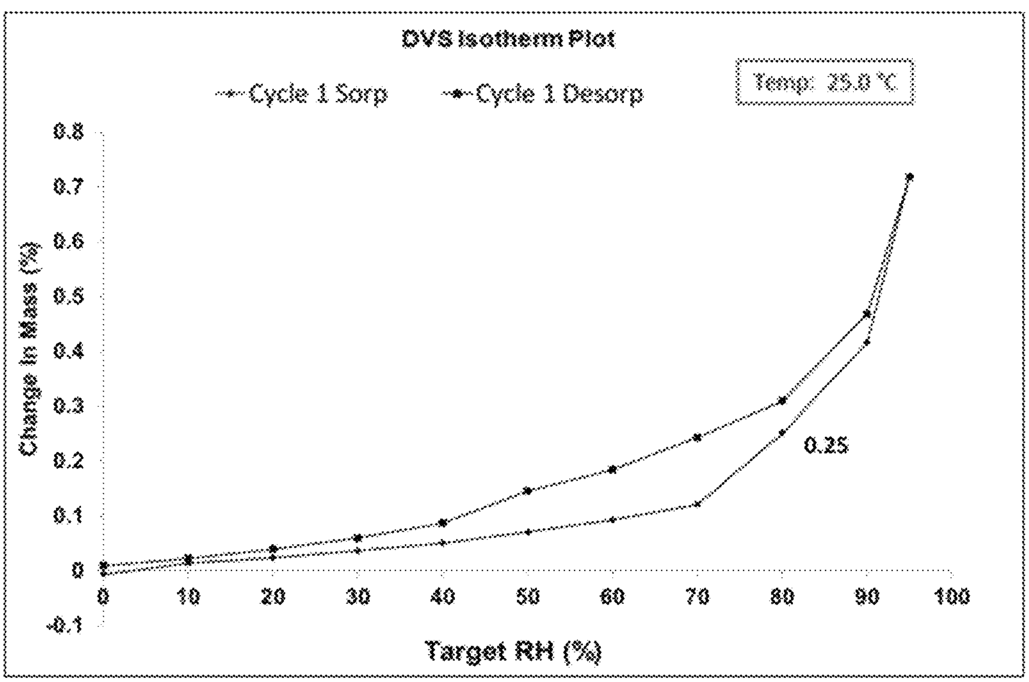
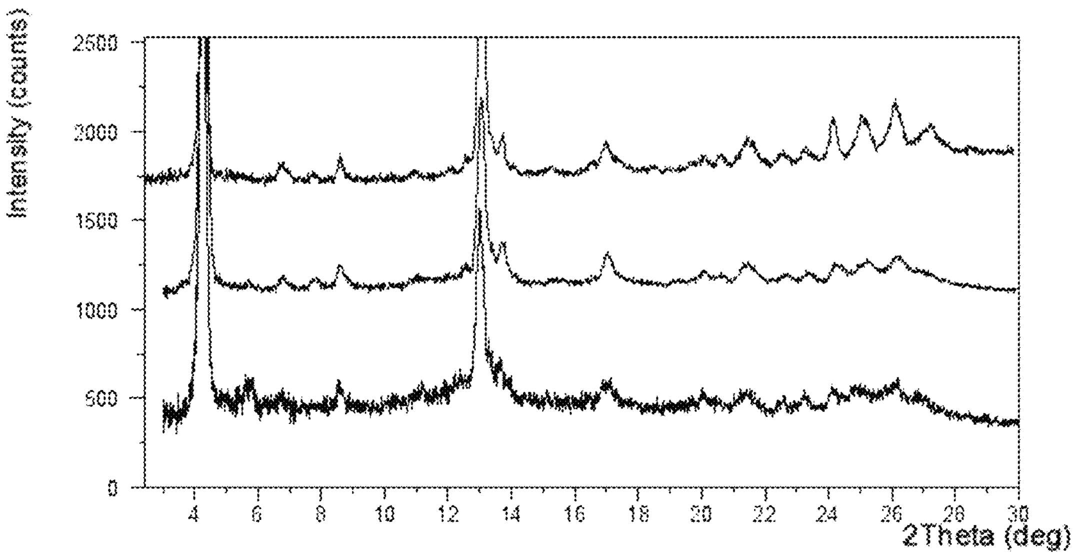

DVS Isotherm Plot
Cycle 1 Sorp
Cycle 1 Desorp
Temp: 25.0 °C
Change In Mass (%)
Target RH (%)
0.25
-0.1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0
10
20
30
40
50
60
70
80
90
100

FIG.26


Intensity (counts)
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1500
1000
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8
10
12
14
16
18
20
22
24
26
28
30
2Theta (deg)

FIG.34

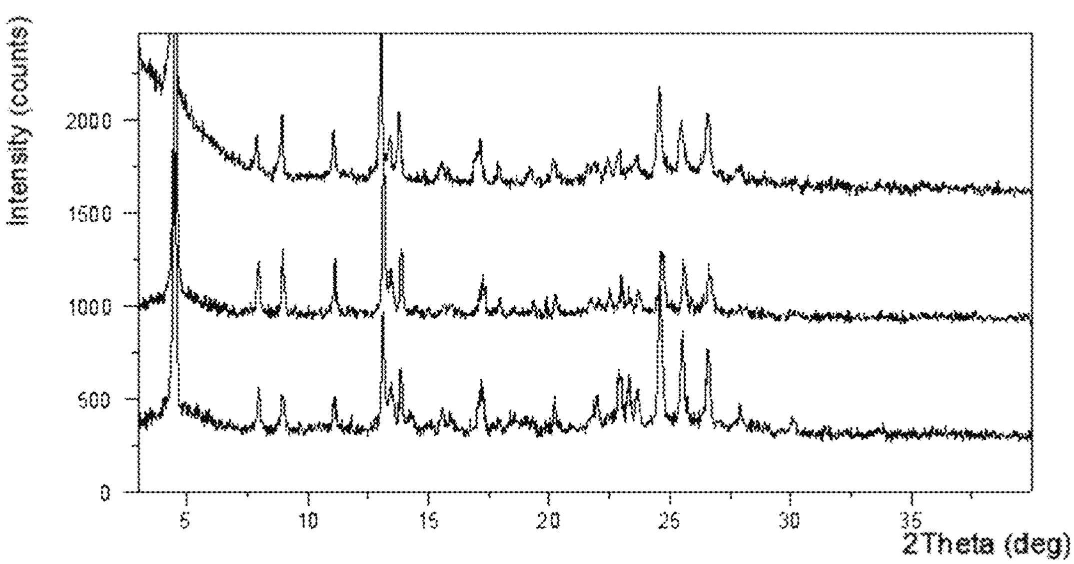

Intensity (counts)
2000
1500
1000
500
0
5
10
15
20
25
30
35
2Theta (deg)

FIG.35

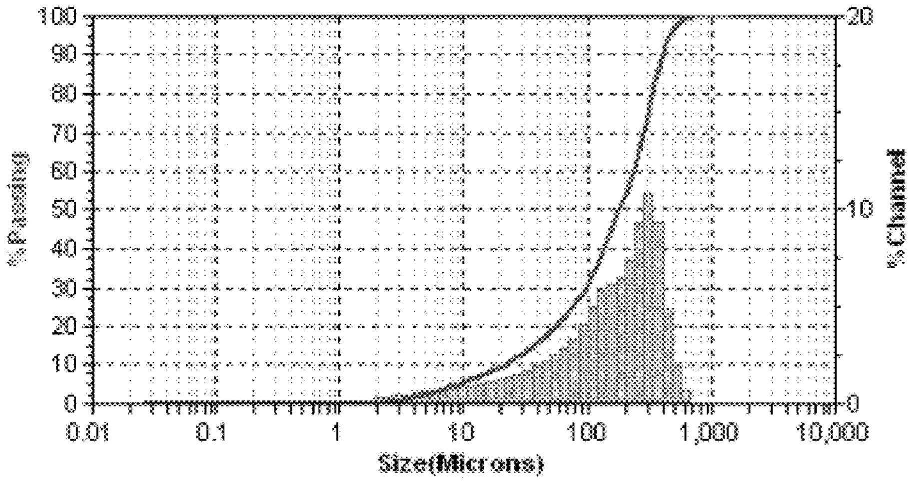


FIG.42

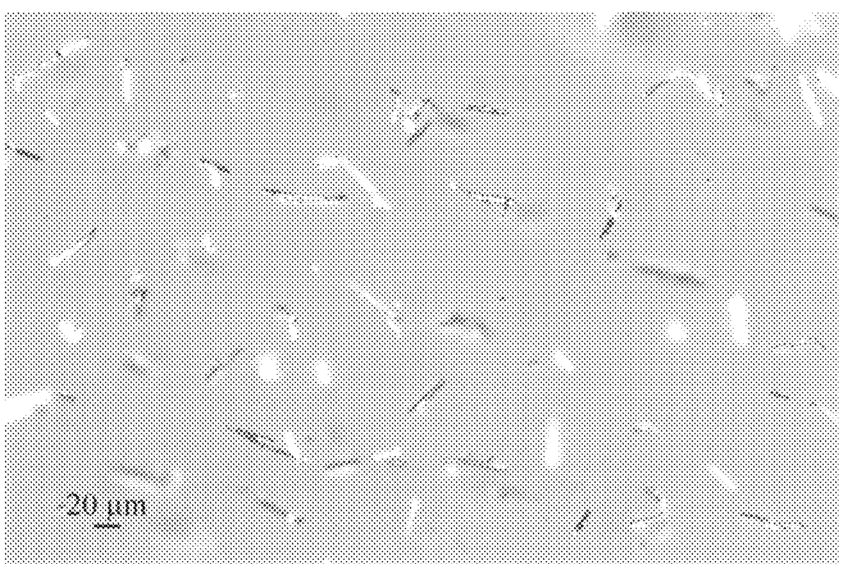


FIG.43

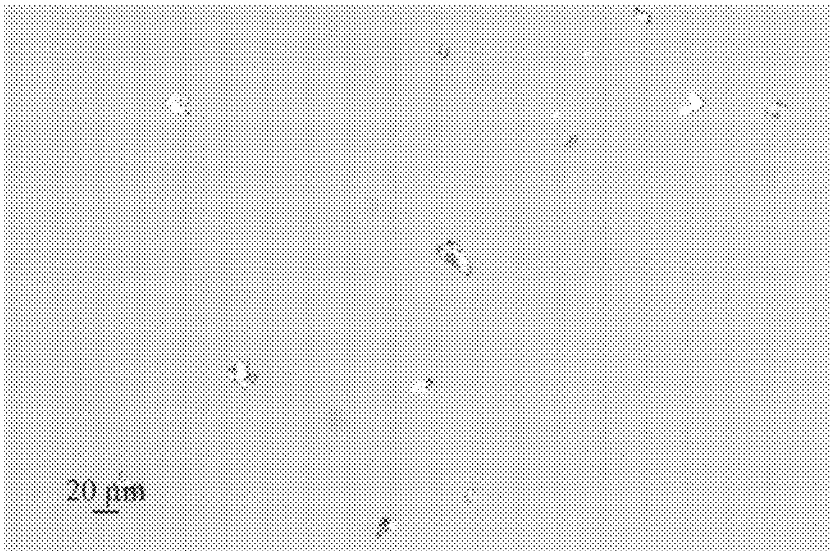


FIG.44

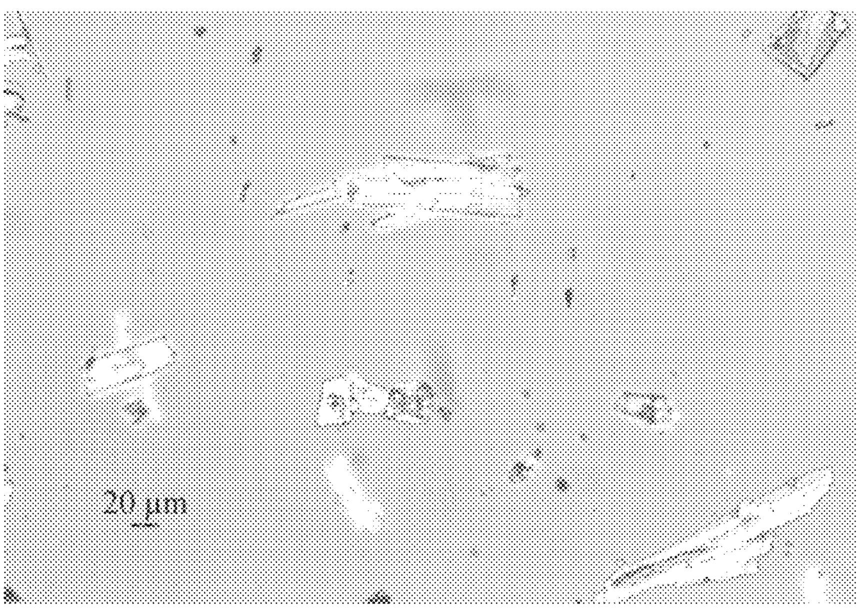


FIG.45

CRYSTALLINE FORMS OF OZANIMOD AND OZANIMOD HYDROCHLORIDE, AND PROCESSES FOR PREPARATION THEREOF

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 16/310,328, filed on Dec. 14, 2018, which is National Stage Entry of International Application No. PCT/CN2017/088314, filed on Jun. 14, 2017, which, in turn, claims the benefit of foreign priority of Chinese Patent Application No.: 201610493910.7, filed on Jun. 29, 2016, and Chinese Patent Application No.: 201610416179.8, filed on Jun. 14, 2016. The entire contents of the aforementioned applications are incorporated herein by reference.

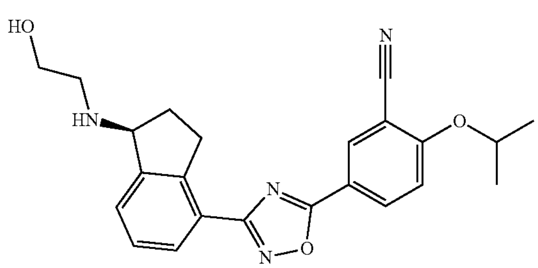
TECHNICAL FIELD

The present disclosure relates to the field of pharmaceutical chemistry, particularly relates to crystalline forms of ozanimod and ozanimod hydrochloride, and processes for preparation thereof.

BACKGROUND

Ozanimod is a novel, oral, selective modulator of sphingosine-1-phosphate 1 receptor (S1P1R) developed by Receptos for the treatment of autoimmune diseases. ozanimod is in phase III trials in the USA for the treatment of multiple sclerosis (MS) and ulcerative colitis (UC). Ozanimod has very excellent pharmacokinetics, pharmacodynamics and safety data in clinical trials, which can perfectly meet the differentiated development strategy and is expected to be the best second-generation SIP1R modulator drug. The chemical structure of the drug is shown as formula

(I)



A specific pharmaceutical activity is the basic prerequisite to be fulfilled by a pharmaceutically active agent to be approved as a medicament on the market. However, there are a variety of additional requirements that a pharmaceutically active agent has to comply with. These requirements are based on various parameters which are connected with the nature of the active substance itself. Without being restricted, examples of these parameters are the chemical stability, solid-state stability and storage stability of the active substance under various environmental conditions, stability during production of the pharmaceutical composition and the stability of the active substance in the final drug products, etc.

The pharmaceutically active agent used for preparing the pharmaceutical compositions should be as pure as possible and its stability in long-term storage must be guaranteed under various environmental conditions. It is essential to prevent using pharmaceutical compositions which contain agent other than the actual active agent, for example decomposition products thereof. In such cases the content of active agent in the drug might be less than that specified. In addition it is important that the pharmaceutically active agent should be non-hygroscopic, stable both to degradation and subsequent solid form changes. If the pharmaceutically active agent is hygroscopic in the sense that it absorbs water (either slowly or over time), it is almost impossible to reliably formulate the pharmaceutically active agent into a drug as the amount of agent to be added to provide the same dosage will vary greatly depending upon the degree of hydration. Furthermore, variations in hydration or solid form can lead to changes in physicochemical properties, such as solubility or dissolution rate, which can in turn lead to inconsistent oral absorption of the patient. Preferably, therefore, a pharmaceutically active substance should be only slightly hygroscopic.

Accordingly, chemical stability, solid-state stability, "shelf life" and materials handling properties (such as ease of solubilizing the compound) of the pharmaceutically active substance are very important factors. In an ideal situation, the pharmaceutically active substance and any compositions containing it should be capable of being effectively stored over appreciable periods of time, without exhibiting a significant change in the physicochemical properties of the active substance such as its activity, moisture content, solubility, solid form, etc. Further, the pharmaceutically active substance usually needs to be processed to achieve a particle size suitable for inhalation and any crystalline form must be stable during such processing so that the properties of the final product are predictable and reliable. In short, in the production of commercially viable and pharmaceutically acceptable pharmaceutical compositions, it is desirable to provide the pharmaceutically active substance in a fully crystalline and stable form, wherever possible.

Different crystalline forms of the same solid chemical drug are significantly different in solubility and stability which can in turn affect the absorption and bioavailability of the drug products. CN102762100A or US2011172202A1 disclosed the compound of formula (I), while there is no solid form or crystalline form of ozanimod disclosed in the prior art. The prior art has neither guidance nor inspiration for finding the crystalline forms. Therefore, it is necessary to perform comprehensive polymorph screening of ozanimod to select a crystalline form which is the most suitable for drug development.

The inventors of the present disclosure have found several crystalline forms of ozanimod and a crystalline form of ozanimod hydrochloride through research process, which provides new choices for the preparation of ozanimod drug product.

SUMMARY

The present disclosure provides novel crystalline forms of ozanimod and ozanimod hydrochloride, and processes for preparation and use thereof.

One objective of the present disclosure is to provide a crystalline form of ozanimod, designated as Form CS1.

The X-ray powder diffraction pattern of Form CS1 shows characteristic peaks at 2theta values of 12.1°±0.2°, 10.4°±0.2° and 4.2°±0.2° using CuKα radiation. Furthermore, the X-ray powder diffraction pattern of Form CS1 shows one or two or three diffraction peaks at 2theta values

Example 30

Solubility Assessment:

Form CS3 and Form CS5 were suspended into SGF (Simulated gastric fluids) and water, and Form CS2 was suspended into FeSSIF (Fed state simulated intestinal fluids) to obtain saturated solutions. After being equilibrated for 1 h, 4 h and 24 h, concentrations of the saturated solutions were measured by HPLC. The results were listed in Table 28.

TABLE 28

	Solubility (mg/mL)				
	FeSSIF	SGF		H_2O	
Time (h)	Form CS2	Form CS3	Form CS5	Form CS3	Form CS5
1	2.0	0.54	1.4	0.17	0.24
4	2.2	0.52	1.0	0.24	0.26
24	1.1	0.54	0.77	0.35	0.27

Example 31

Mechanical Stability of Form CS3, Form CS5, Form CS6 of Ozanimod and Form CS1 of Ozanimod Hydrochloride:

Solid samples of Form CS3, Form CS5, Form CS6 of ozanimod and Form CS1 of ozanimod hydrochloride were ground manually for 5 minutes in mortar. The XRPD patterns were displayed in FIG. **49**, FIG. **50**, FIG. **51** and FIG. **52**, respectively (the top pattern is before grinding and bottom pattern is after grinding).

No form change was observed for Form CS3, Form CS5, Form CS6 of ozanimod and Form CS1 of ozanimod hydrochloride under a certain mechanical stress with slightly decrease in crystallinity. Good physical and chemical properties including stability made these forms suitable for drug preparation and storage.

Crystalline forms with better mechanical stability have good physicochemical properties and remain stable under certain mechanical stress. The crystalline drug with better mechanical stability has low requirements on the crystallization equipment, and no special post-treatment condition is required. It is more stable in the formulation process, can significantly reduce the development cost of the drug products, enhance the quality of the drug, and has strong economic value.

It is to be noted that Form CS1 used in Examples 16 to 31 of the present disclosure is prepared by the method of Example 1; Form CS2 is prepared by the method of Example 3; Form CS3 was prepared by the method of Example 6; Form CS5 was prepared by the method of Example 9; Form CS6 was prepared by the method of Example 10; Form CS1 of ozanimod hydrochloride was prepared by the method of Example 11.

The examples described above are only for illustrating the technical concepts and features of the present disclosure, and intended to make those skilled in the art being able to understand the present disclosure and thereby implement it, and should not be concluded to limit the protective scope of this disclosure. Any equivalent variations or modifications according to the spirit of the present disclosure should be covered by the protective scope of the present disclosure

The invention claimed is:

1. A crystalline Form CS1 of ozanimod hydrochloride, wherein the X-ray powder diffraction pattern shows characteristic peaks at 2theta values of 26.1°±0.20°, 24.4°±0.20° and 20.1°±0.20° using CuKα radiation.

2. The crystalline Form CS1 of ozanimod hydrochloride according to claim **1**, wherein the X-ray powder diffraction pattern shows 1 or 2 or 3 characteristic peaks at 2theta values of 3.9°±0.20°, 21.1°±0.20° and 7.9°±0.20° using CuKα radiation.

3. The crystalline Form CS1 of ozanimod hydrochloride according to claim **1**, wherein the X-ray powder diffraction pattern shows 1 or 2 or 3 characteristic peaks at 2theta values of 11.9°±0.20°, 19.6°±0.20° and 13.8°±0.20° using CuKα radiation.

4. A process for preparing crystalline Form CS1 of ozanimod hydrochloride according to claim **1**, wherein the process comprises method 1) or method 2) or method 3) or method 4) or method 5),

1) adding ozanimod hydrochloride into an ether and stirring at 4-50° C., filtering and drying to obtain a white solid of Form CS1 of ozanimod hydrochloride; said stirring time is at least 0.5 hour; or
2) dissolving ozanimod hydrochloride into a mixture of an alcohol and an ester, evaporating at room temperature to obtain a white solid of Form CS1 of ozanimod hydrochloride; said evaporating time is at least 0.5 day; or
3) dissolving ozanimod hydrochloride into a solvent selected from an amide or a mixture of solvents thereof, then placing the solution in a system containing an antisolvent of ozanimod hydrochloride for liquid vapor diffusion at room temperature, filtering and drying to obtain a white solid of Form CS1 of ozanimod hydrochloride; said diffusion time is at least 1 day; or
4) dissolving ozanimod hydrochloride into a mixture of an alcohol and water to form a supersaturated solution, ozanimod hydrochloride is fully dissolved at 25-80° C. and then filtering, cooling the filtrate for precipitation, filtering and drying to obtain a white solid of Form CS1 of ozanimod hydrochloride; or
5) dissolving ozanimod hydrochloride into an alcohol in a concentration of 12 mg/mL to form an alcohol solution, filtering and evaporating the alcohol solution at room temperature for 1 week to obtain a white solid of Form CS1 of ozanimod hydrochloride.

5. A pharmaceutical composition, wherein said pharmaceutical composition comprises a therapeutically effective amount of crystalline Form CS1 of ozanimod hydrochloride according to claim **1** and a pharmaceutically acceptable carrier, a diluent or an excipient.

6. A method of treating ulcerative colitis, comprising administering to a patient in need thereof a therapeutically effective amount of crystalline Form CS1 of ozanimod hydrochloride according to claim **1**.

7. A method of treating multiple sclerosis, comprising administering to a patient in need thereof a therapeutically effective amount of crystalline Form CS1 of ozanimod hydrochloride according to claim **1**.

* * * * *